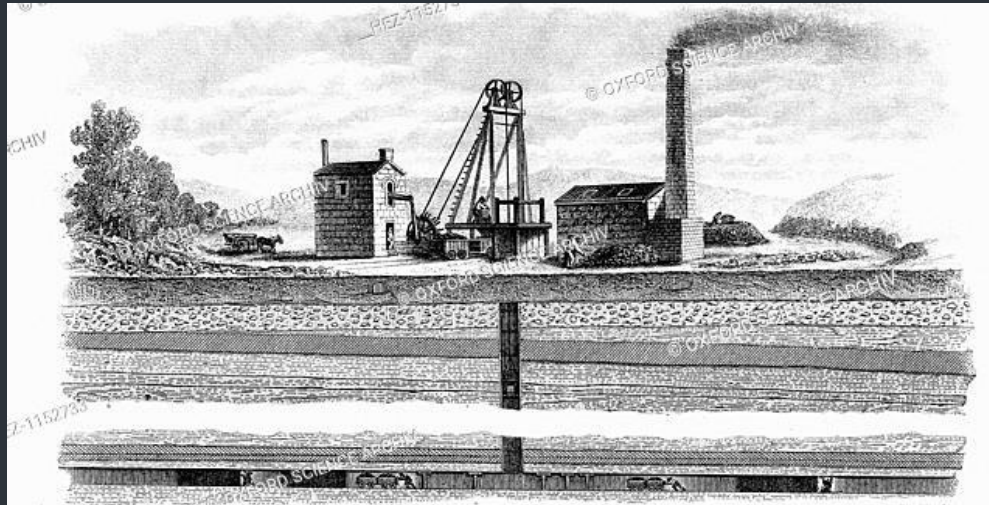


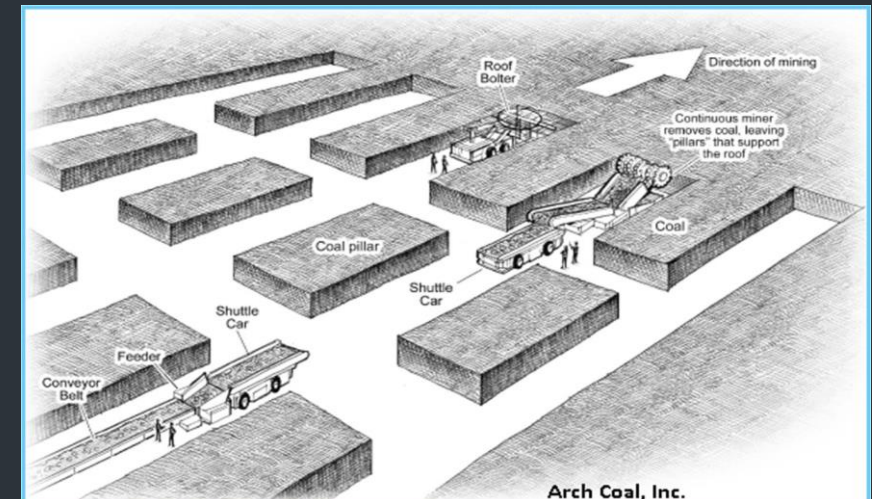
Underground Coal Mine Planning



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Contents

- ▶ Optimization of mine size (mine production capacity) based on techno-economic considerations;
- ▶ Equipment and face scheduling against targeted production using suitable software.



Introduction

- ▶ Coal is the most abundant fossil fuel on Earth. Coal is found in beds or seams interstratified with shales, clays, sandstones, or (rarely) limestone.
- ▶ In underground coal mining, the working environment is completely enclosed by the geologic medium, which consists of the coal seam and the overlying and underlying strata. Access to the coal seam is gained by suitable openings from the surface, and a network of roadways driven in the seam then facilitates the installation of service facilities for such essential activities as human and material transport, ventilation, water handling and drainage, and power.
- ▶ The most economical method of coal extraction from coal seams depends on the depth and quality of the seams, and the geology and environmental factors. Many coal deposits are extracted from both surface and underground mines. Surface mining and deep underground mining are the two basic methods adopted for mining coal. The choice of mining method depends primarily on depth of burial, density of the overburden and thickness of the coal seam.



Development process of a new mine

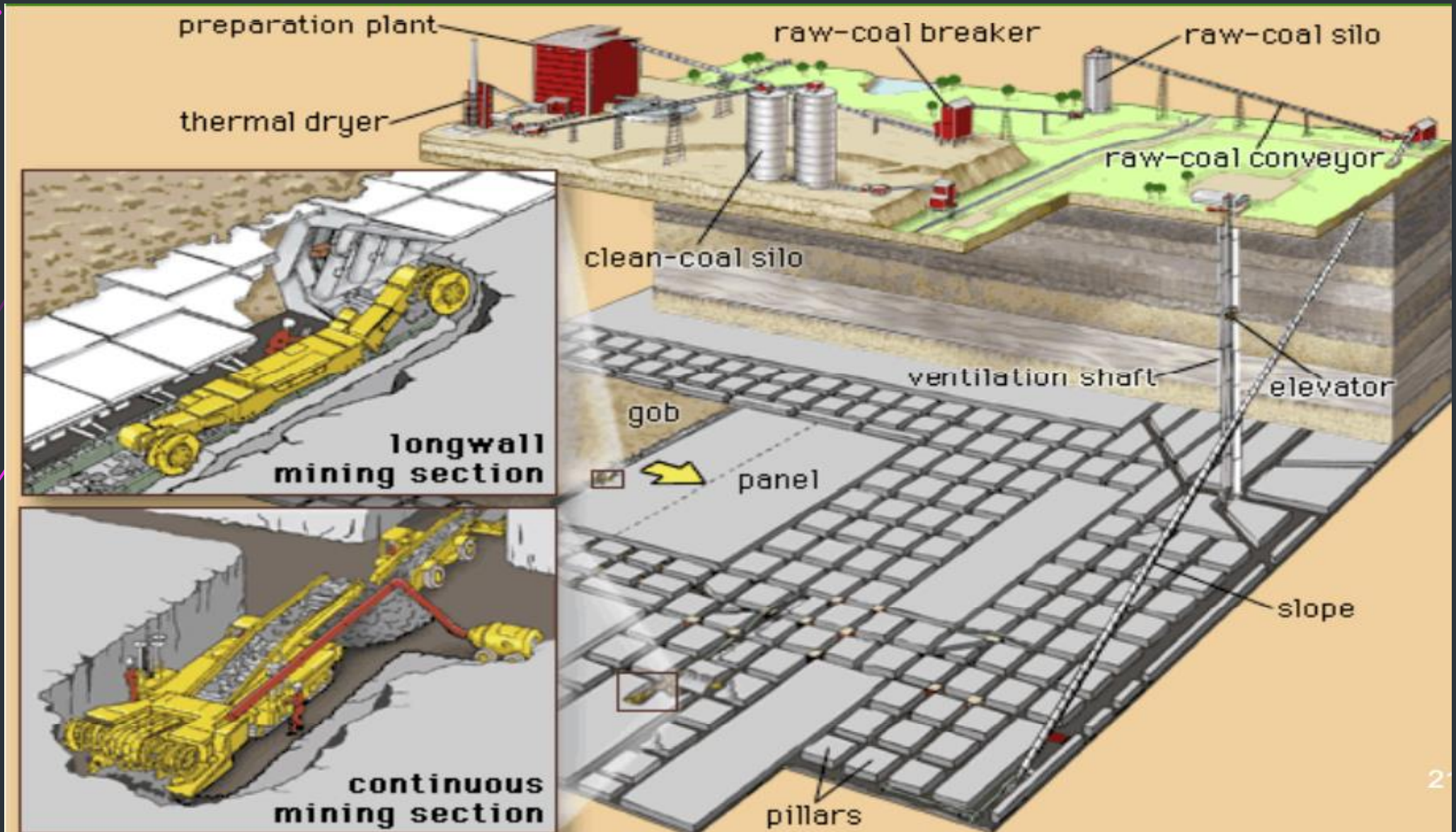
- Geological survey
- Exploration
- Mine development : Incline/shaft
- Mining: B&P, LW/SW
- Rehabilitation: erosion control, revegetation etc.
- Environmental monitoring



Mining methods

- ▶ Surface/open cut mines:
 - ▶ Strip mining; dragline
 - ▶ open pit mining; truck & shovel, BWE etc BWE etc.
- ▶ Underground mines: manual, semi mechanized, fully mechanized
 - ▶ Bord and-pillar mining
 - ▶ Longwall/shortwall mining
- ▶ Other methods depending on seam conditions

UNDERGROUND MINING SYSTEM

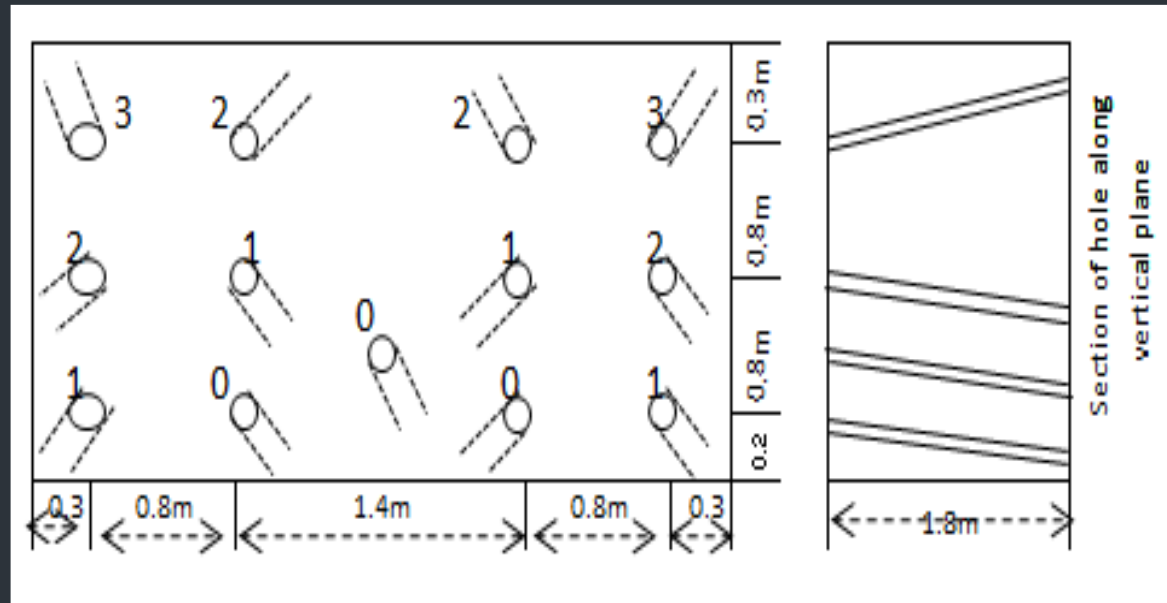


CMR-2017,Rule-111

Depth of seam from surface	Where the width of the galleries does not exceed 3.0 meters	Where the width of the galleries does not exceed 3.6 meters	Where the width of the galleries does not exceed 4.2 meters	Where the width of the galleries does not exceed 4.8 meters
The distance between centres of adjacent pillars shall not be less than				
(1)	(2)	(3)	(4)	(5)
	Meters	Meters	Meters	Meters
Not exceeding 60 meters	12.0	15.0	18.0	19.5
Exceeding 60 but not exceeding 90 meters	13.5	16.5	19.5	21.0
Exceeding 90 but not exceeding 150 meters	16.5	19.5	22.5	25.5
Exceeding 150 but not exceeding 240 meters	22.5	25.5	30.5	34.5
Exceeding 240 but not exceeding 360 meters	28.5	34.5	39.5	45.0
Exceeding 360 meters	39.0	42.0	45.0	48.0

Development by drilling and blasting

- A hand held coal drill of 38 mm diameter is being used to drill blast holes of 6 feet length. The drilling pattern is wedge cut in a gallery size of 3.6m x 2.1m. A total of 14 holes per round are drilled for the blasting of the development face.
- Cartridged permitted explosives (32mm in diameter, weighing 185gm)/P5
- The mean depth of blast holes, in various rounds of solid blasts, in general, was 1.8m. The charge per round varied from 4.81 kg to 9.62 kg and the maximum charge per delay varied from 1.48 kg to 3.33 kg.

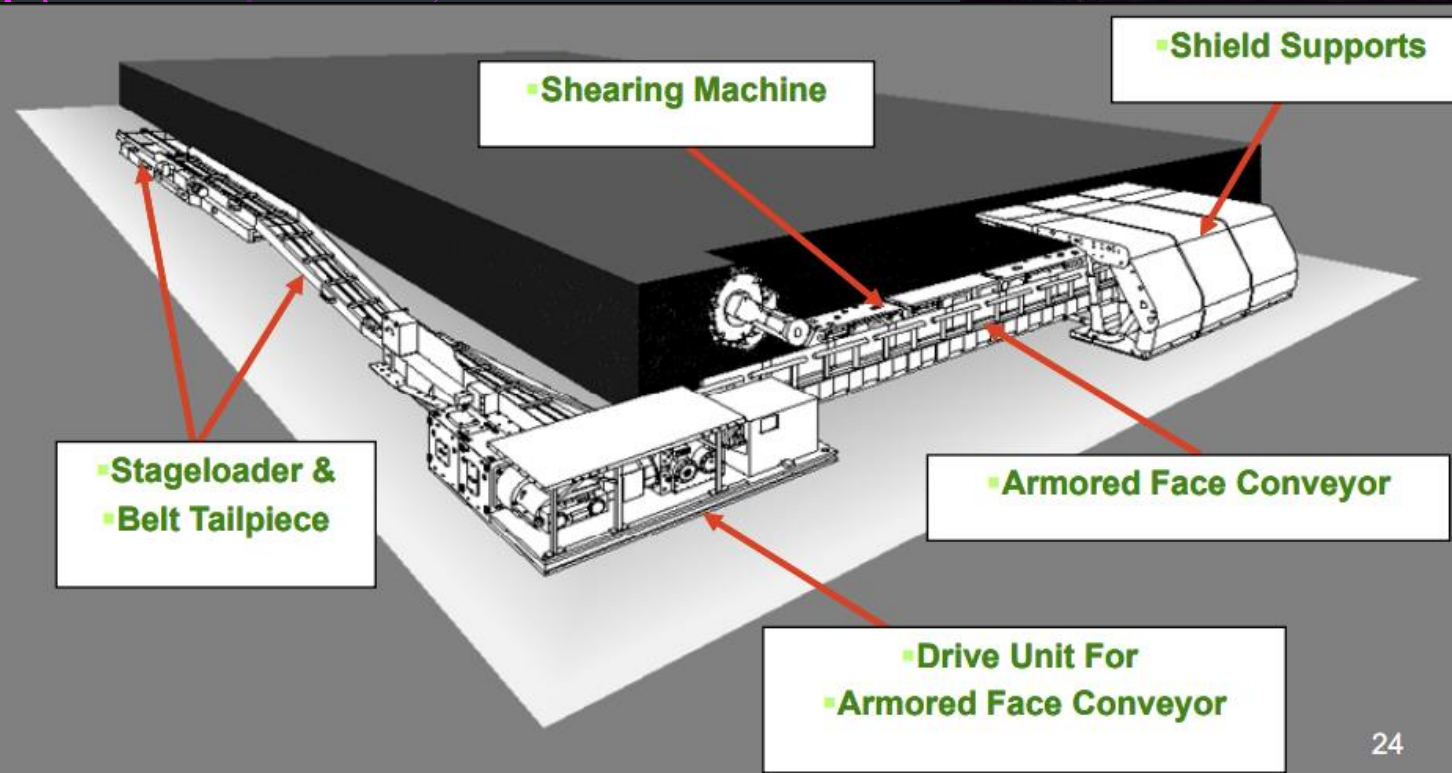




BORD AND PILLAR BY CONTINUOUS MINER AND SHUTTLE CAR



22



LONGWALL SYSTEM

24

Exploration

SG-4									
RUN	EXTRAPOLATED		SEAM	NO.	PROXIMATE ANALYSIS (AS ANALYSED BASIS)				
	DEPTH	THICK			MOIST	ASH	V.M	V.M	
m	m	m			%	%	%	%	
257.70									
258.70									
259.70	859.94								
	260.29 0.35		IA		5.4	25.3	28.6	40.7	
	260.64 0.25				2.0	73.4			
	260.90 0.16				2.9	60.6			
250.75	260.91 0.11				0.9	56.5	24.9	36.6	
	261.02 0.17				4.6	27.9			
	261.27 0.06				4.9	32.7	25.0	37.4	
261.76	261.27 0.48		IB						
	261.75								
	0.96				6.4	29.0	24.7	39.9	
263.65	262.44 0.08		IC		5.6	37.2			
	262.62 0.62				7.4	23.7	28.9	40.0	
	262.14								
269.90									
	270.58 0.20				4.3	43.7			
	270.78 0.84		IIA		4.6	21.0	29.6	44.8	
	271.62								
272.45	272.45								
273.05	273.05		IIIB		4.2	17.6	31.9	46.3	
	273.05								
274.90									
275.90									
	276.73 0.99				3.0	58.1	26.4	39.5	
	277.02 0.27				4.6	21.7	28.7	45.4	
	277.42 0.87				4.2	40.0			
	277.82 0.87				3.7	22.7	28.9	44.7	
278.00	278.00 0.07		III		3.6	51.7			
	278.00								
			IVA						
296.20									
	297.67 0.10				4.9	18.8	29.7	45.6	
	298.09 0.11				4.9	42.5			
	1.31		IVB		5.6	18.3	28.7	46.9	
299.40	299.40								
215.20									
	316.54 0.44								
	317.30 0.05								
317.30	318.13		VA		3.4	37.6			
318.50	318.46								
320.30									

SG-105									
RUN	EXTRAPOLATED		SEAM	NO.	PROXIMATE ANALYSIS (AS ANALYSED BASIS)				
	DEPTH	THICK			MOIST	ASH	V.M	V.M	
m	m	m			%	%	%	%	
275.36									
276.75	276.75		IC	0.85	3.9	27.2	27.3	41.0	
277.60	277.60								
281.85	281.85		IIA	0.24	NOTE	ANALYSED			
	282.02								
	282.26								
	282.86		IIIB	0.60	3.6	27.8	20.5	38.1	
	283.26			0.39	N	S	S		
283.45									
285.00	285.16			0.98	N	S	S		
	286.13				3.4	30.1	27.6	38.9	
	286.15		III	0.38	N	S	S		
	286.66			0.15	N	S	S		
	286.99			0.33	4.5	29.5	22.0	44.0	
287.65									
305.55									
305.55									
	306.90 0.27				N	S	S		
	307.17								
	308.04		IVB	0.87	3.7	17.7	31.3	47.3	
308.55									
309.55									
310.46									
511.65	311.55		IVB	1.09	1.2	30.4	26.4	40.0	
	314.30 0.42				2.6	54.6			
	316.54 0.44								
317.30	317.30 0.05				N	S	S		
					2.8	39.2			

SG-113									
RUN	EXTRAPOLATED		SEAM	NO.	PROXIMATE ANALYSIS (AS ANALYSED BASIS)				
	DEPTH	THICK			MOIST	ASH	V.M	F.C	
m	m	m			%	%	%	%	
180-25									
	182-72 0.05		IA		1.3	52-0			
	0.62				2.3	26.8	29.3	41.6	
183-40	182-74 0.11		IA NW		N	S	S		
	183.83 0.18				2-2	37-2	25-5	35-1	
	184.08 0.25								
	0.91		IB NW		3-0	34-4	25-1	37-5	
185.22	185.22 0.05				2-3	37.8			
	186.18 0.08								
	186.82 0.06				2.7	37-3			
	186.73 0.36				1.8	66.4			
	0.61				2.7	29.1	26.4	41.8	
	186.99 0.26				2.2	46.3			
188.09			IC W		2.5	24.8	29.1	43.6	
	2.30								
189.55									
194.15									
	195.34 0.14				N	S	S		
	195.41 0.15								
	195.56		IIA NW		2.0	27.3	28.4	42.3	
156.53									
197.30	197.30 0.10								
200.05									
	202.11				2.4	22.6	30.	44.5	
	1.20		III W		1.7	59.6	29.4	42.1	
203.55	203.31 0.12				2.2	26.3			
	203.71 0.26								
	202.11								
222.30									
	223.33 0.44				2.5	28.4	26.2	42.9	
	223.56 0.08				2.4	47.6			
	223.71		IVB NW		3.0	17.1	28.3	51.6	
	223.65								
	1.33								
225.80	225.16								

Reserve of the Block

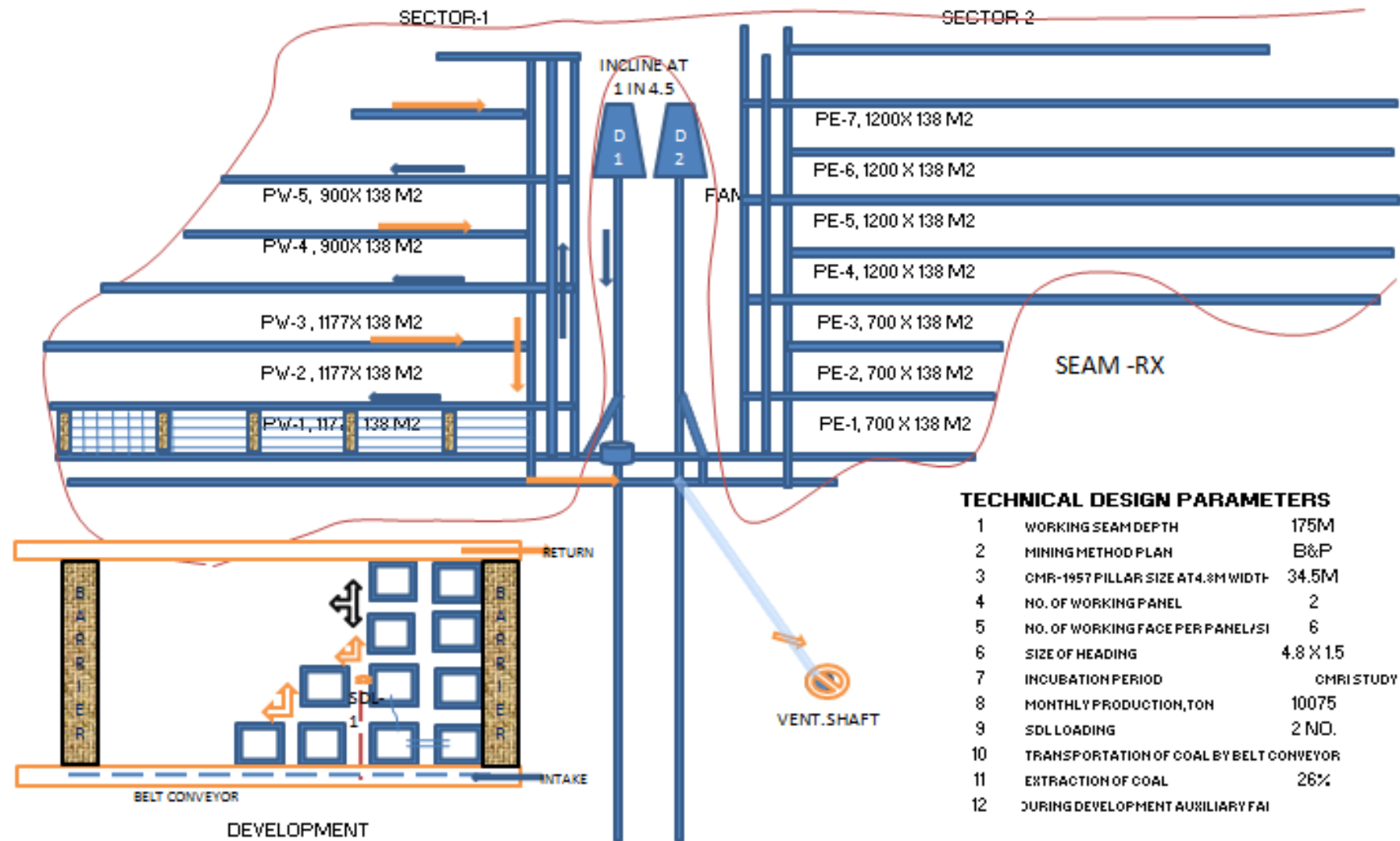
Mineable reserves (Quantity – '000 t)

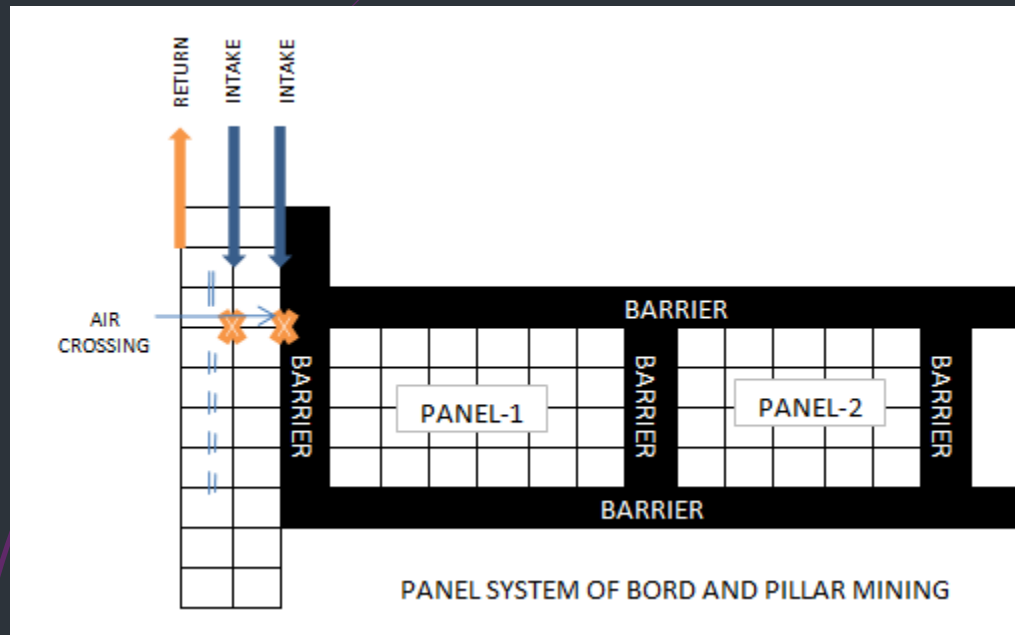
Seam	Thickness Range (m)	Reserves in depth < 250 m	Reserves in depth > 250 m	Total
IC	2.5 – 3.5	4091	8785	12876
III	1.2-2.0	1441	1239	2680
Total		5532(35.56%)	10024(64.44%)	15556

Seam	Extractable Reserves	Recoverable Reserves
	(M. Tonnes)	(M. Tonnes)
IC	7.72	5.02
III	1.34	0.67
Total Reserves	9.06	5.69

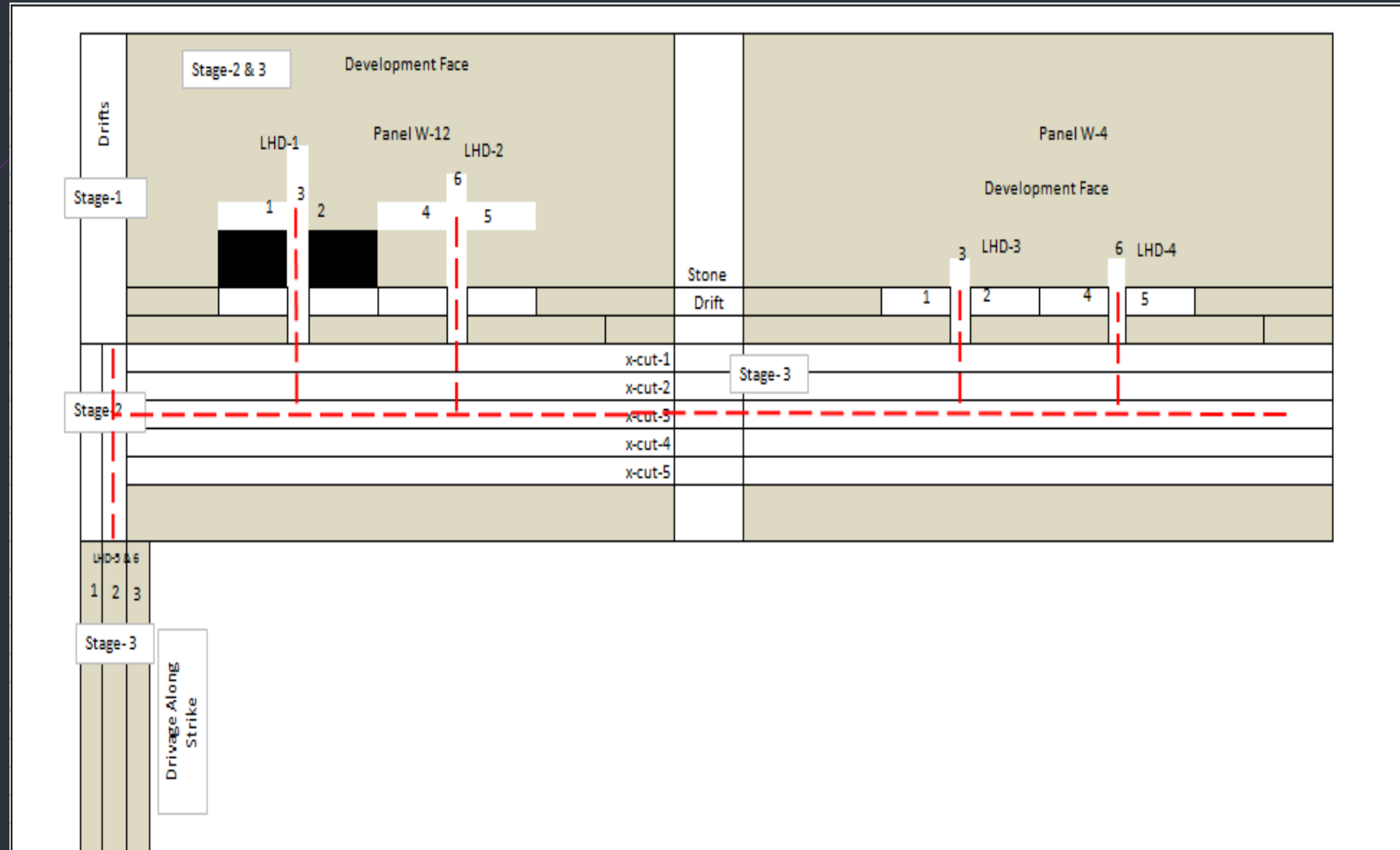
Life of Mine: Reserve of the mine from extractable to recoverable varies from 9.06MT to 5.69MT so it directly depends on the mining efficiency but after all constraints life of mine will be more than 20 years at the production rate of 0.3MT per year.

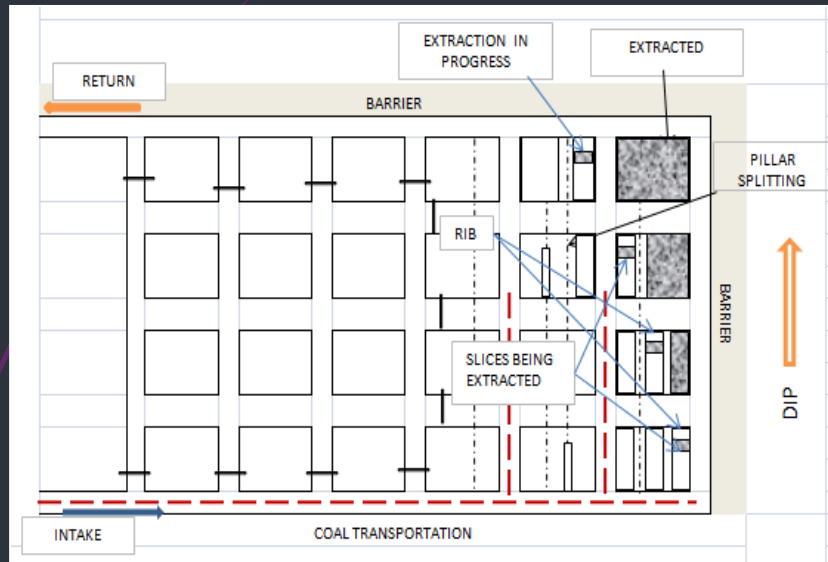
SEAM R-X MINING SEQUENCE BY BORD AND PILLAR MINING METHOD



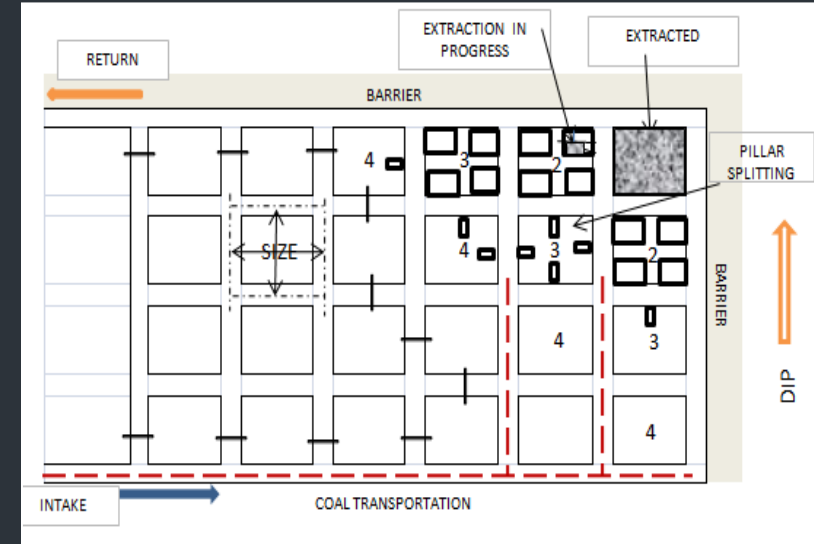


PANEL -1		UNIT
Length	207	M
Width	138	M
Seam Thickness, R-X	1.5	M
Pillar Size (Centre To Centre)	34.5	M
No. Of Pillars	24	
Total Coal Yield	55704	Tons
Production Per Day(Approx)	200	Tons
Production Per Month@25	5000	Tons
Life Of Panels (Approx)	11	Months





Each pillar of 40mx40m centre and 48m x 48m will be split at 1/3rd of pillar width by a level split of 4.2/4.8m width



■ Each pillar of 34.5m x 34.5m centre will be split into two halves by a level split of 4.8m wide.

The target production of 1600 TPD is likely to be achieved in 6th year of operation if all the milestone activities are completed on schedule.

Coal Production Biharinath Coal Block		
Yr 0: Once Box Cut and Portal have been completed this time programme starts with month 0.		
Period	Approved Mining Plan	Possible Production
0 To 24 months	0	0
25 TO 36 months	30,000	150,000
Yr 4	100,000	350,000
Yr 5	260,000	410,000
Yr 6	400,000	585,000
Yr 7	480,000	585,000
Yr 8	480,000	585,000
Yr 9	480,000	585,000
Yr 10	480,000	585,000
Total 10 Yrs	2,710,000	3,835,000
Yr 11 to 20	4,800,000	5,850,000
Yr 21 to 30	4,800,000	5,850,000
Total Project	12,310,000	15,535,000

